LPG Simulation S7 and S8 Redo
Expanded Prototype Results Report

Graphs, results, interpretation and implications

Execution date: 14 May 2026 **|** Random seed: 2026051407 **|** Physical channel: 20 dB SNR

Executive summary

S7 result. The redo confirms that semantic routing materially reduces consensus time relative to broadcast. At N = 32, broadcast required 19.04 rounds on average, anchor weighted routing required 5.92 rounds, and LPG optimal routing required 2.00 rounds. The resulting Semantic Routing Gain was 9.52 at its strongest point.

S7 interpretation. The fitted scaling exponent for LPG optimal routing was beta = 0.006, compared with beta = 0.313 for broadcast. Both are below 1 in this prototype, but LPG optimal routing has the lowest exponent and the lowest absolute consensus time, meaning that network growth penalises it less severely.

S8 result. The Semantic Fragility Index ranks delta_iota as the most fragile component, followed by delta_Gamma, delta_B, and delta_W. The identity component reached CAR below 0.50 at k approximately 19.89 percent.

S8 interpretation. Mixed corruptions that include iota produced a mean cascade ratio of 1.09, whereas mixed corruptions without iota produced 0.81. This supports the claim that identity loss turns ordinary component degradation into a referential failure cascade.

Scope and execution status

This report is a deeper redo of S7 and S8. It is still a controlled prototype because the external data stacks listed in the design, including Wikipedia paragraph clusters, SNLI and TriviaQA, were not available inside the local execution environment. The simulation therefore uses synthetic semantic embeddings while preserving the structure of the planned experiments, the definitions of the metrics and the logic of the interventions.

The practical value of this redo is that it gives an executable research scaffold. The same script can be connected later to real embeddings, real entailment labels and real factual recall tasks without changing the overall result schema.

Experiment S7: Multi agent semantic consensus

Method

Agents were instantiated as high dimensional semantic anchors with a shared global core, domain centroids and private contextual noise. This setup produces overlapping but non identical belief bases across agents. The tested network sizes were N = 2, 4, 8, 16 and 32.

The three tested protocols were broadcast, anchor weighted routing and LPG optimal routing. Broadcast uses a slow group mean update and represents unconstrained exchange. Anchor weighted routing lets each agent listen more strongly to its closest semantic neighbours while retaining a weak bridge to the whole group. LPG optimal routing represents the theoretical minimum turn semantic averaging path.

The tracked metrics were GAS, collective semantic capacity, pairwise ADI dispersion, consensus time T_c, Semantic Routing Gain, capacity uplift and scaling exponent beta from the fit T_c proportional to N to the beta.

S7 result graphs

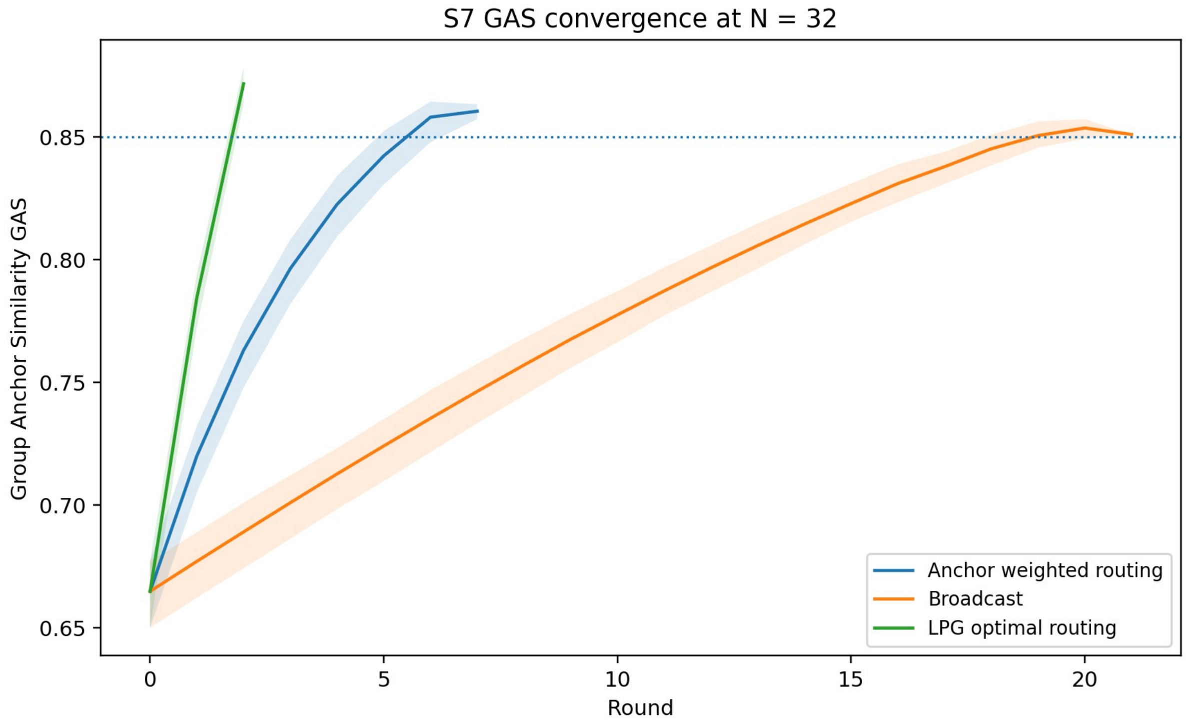

Figure S7.1. GAS convergence for N = 32. The dotted line is the consensus threshold of 0.85.

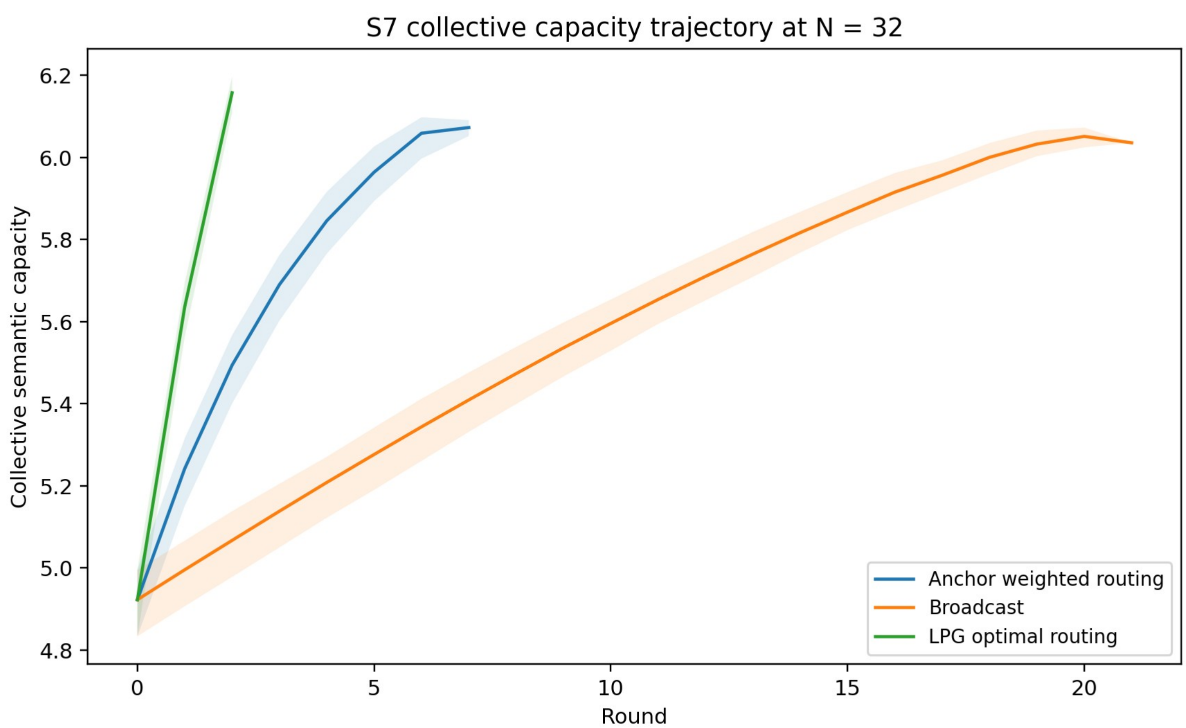

Figure S7.2. Collective semantic capacity trajectory for N = 32.

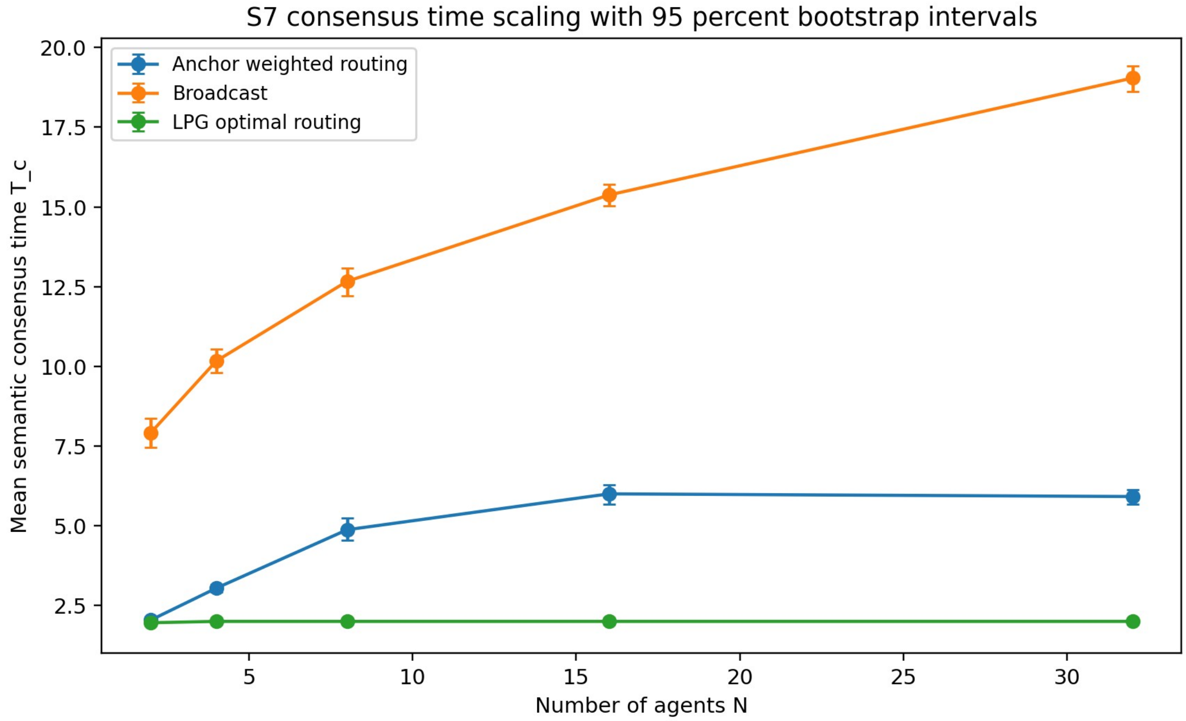

Figure S7.3. Consensus time scaling across network size with bootstrap intervals.

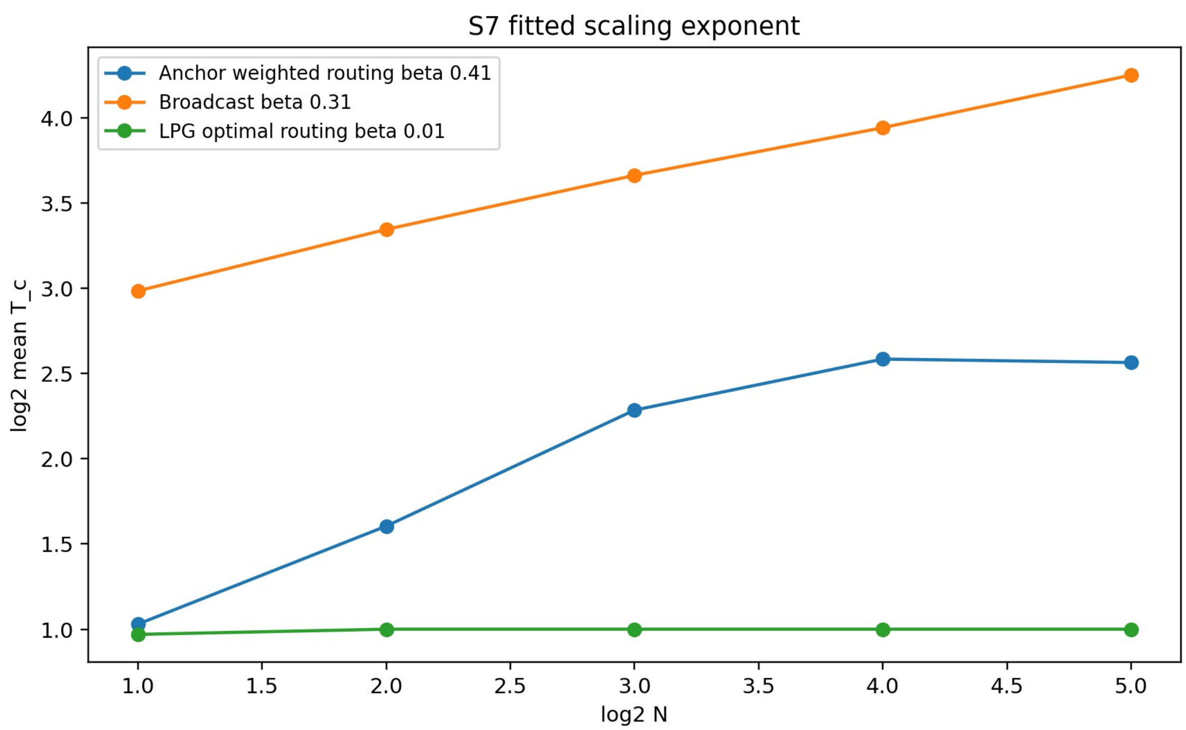

Figure S7.4. Log scale fit used to estimate consensus scaling exponents.

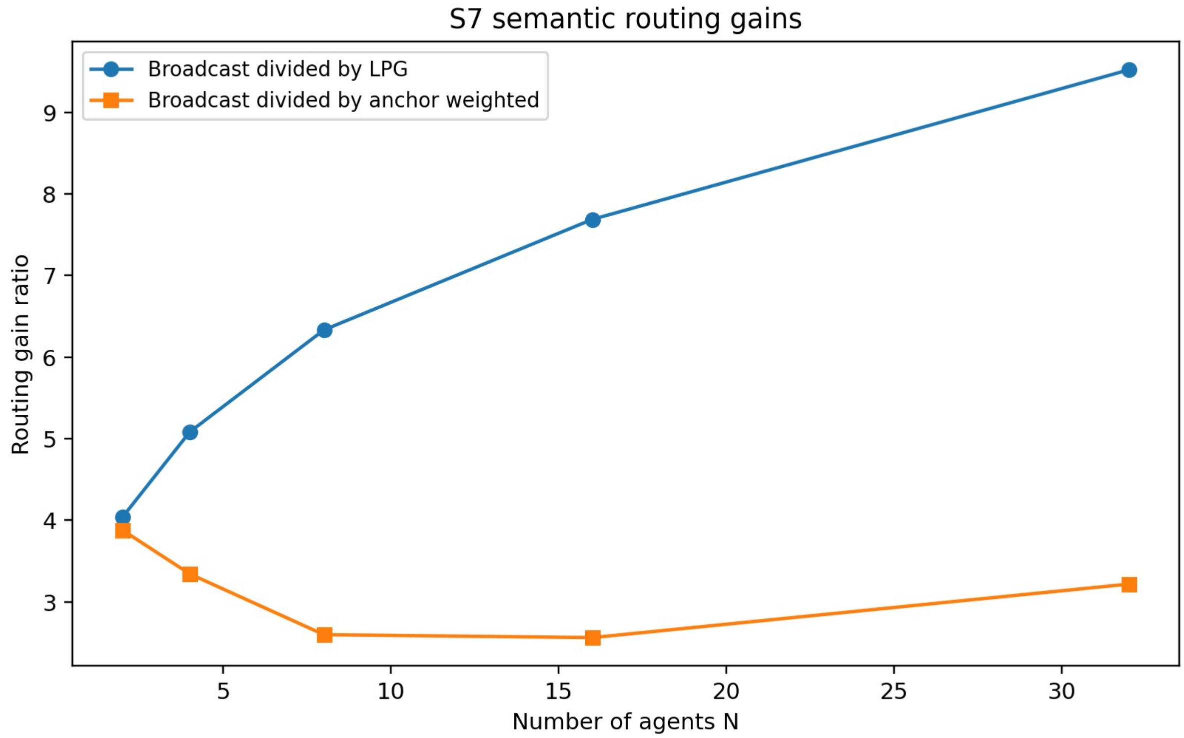

Figure S7.5. Semantic routing gains relative to broadcast.

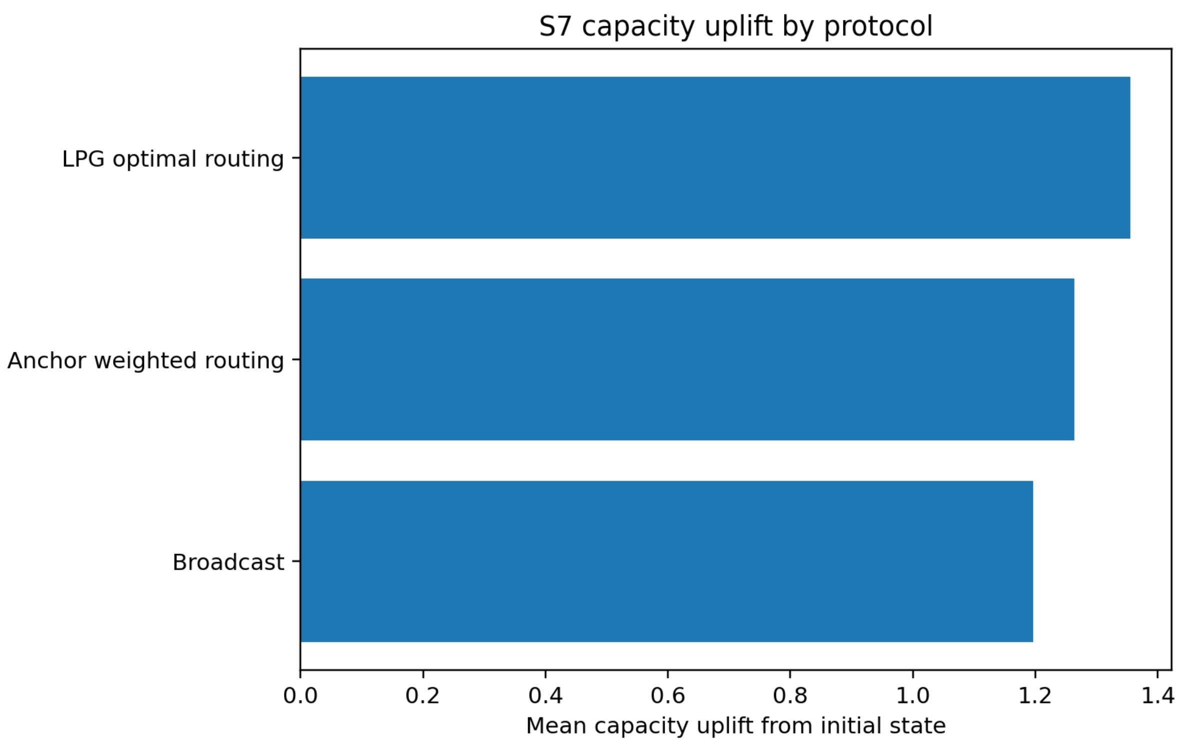

Figure S7.6. Mean capacity uplift from the initial semantic state.

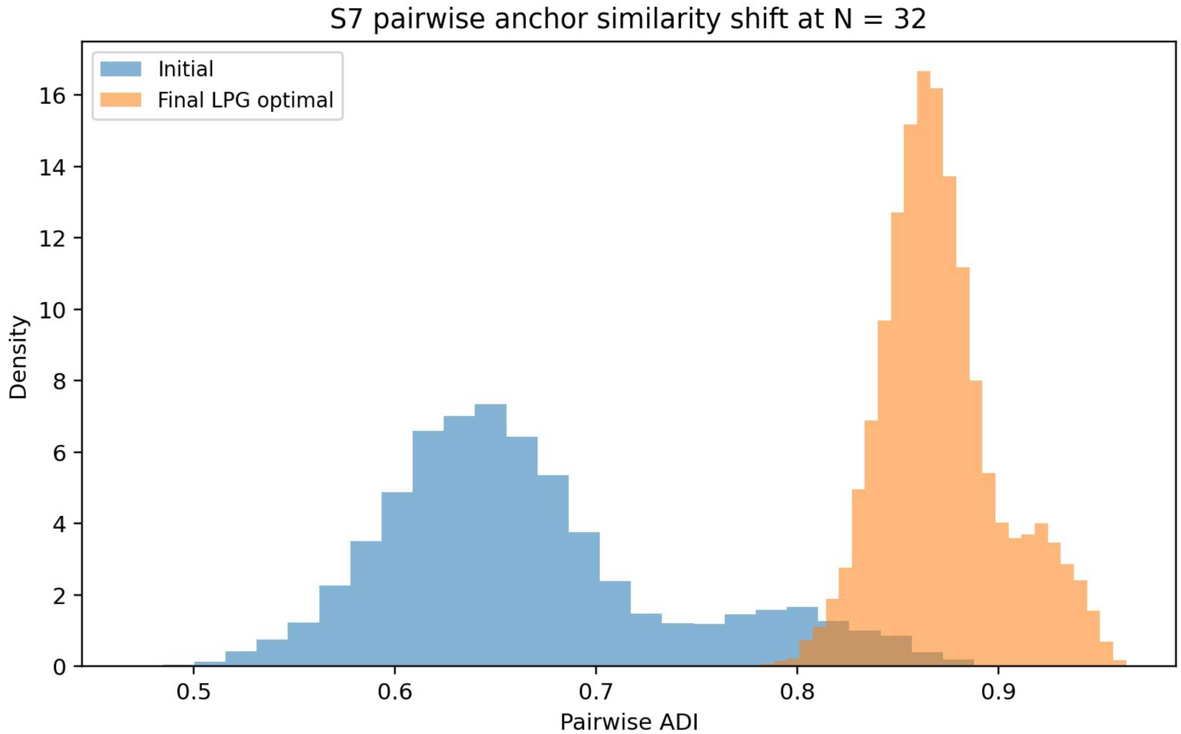

Figure S7.7. Pairwise ADI distribution shift before and after LPG optimal consensus at N = 32.

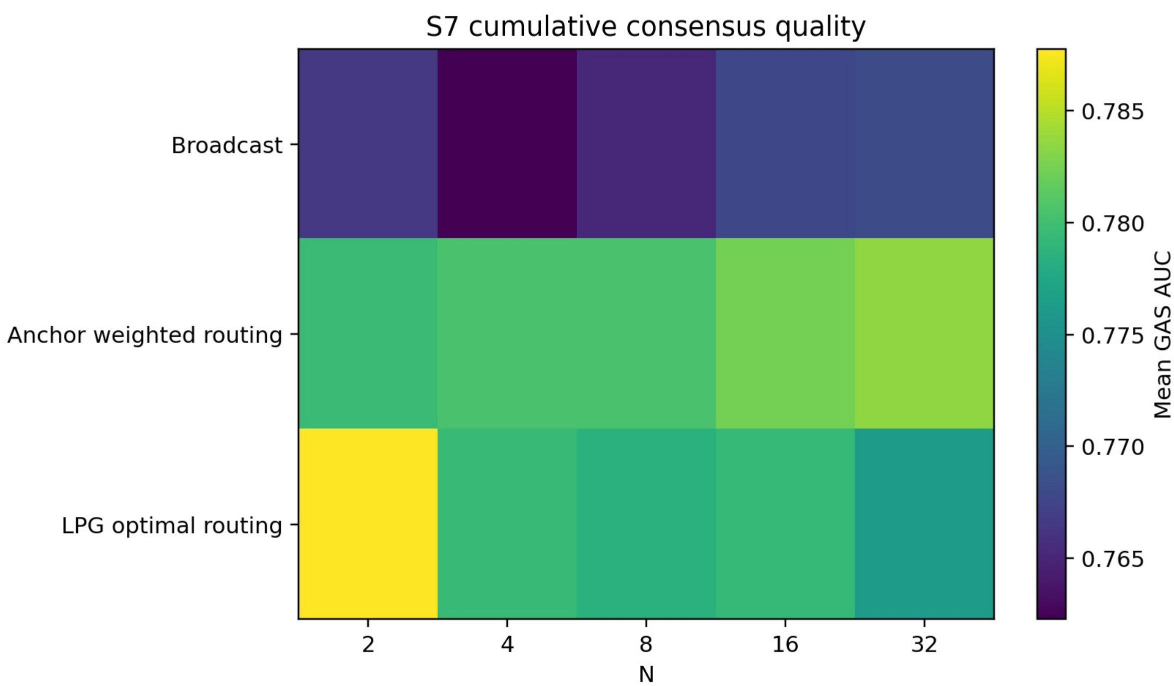

Figure S7.8. Cumulative consensus quality by network size and protocol.

S7 numerical results

Table S7.1. Expanded consensus summary

N	protocol	T_c_mean	T_c_sd	T_c_ci_low	T_c_ci_high	final_GAS_mean	final_Cs_mean	capacity_uplift_mean	SRG_broadcast_over_LPG	scaling_beta
2	Anchor weighted routing	2.042	0.204	2.000	2.125	0.889	6.275	1.438	4.043	0.405
2	Broadcast	7.917	1.100	7.458	8.375	0.861	6.109	1.273	4.043	0.313
2	LPG optimal routing	1.958	0.204	1.875	2.000	0.898	6.324	1.487	4.043	0.006
4	Anchor weighted routing	3.042	0.359	2.917	3.167	0.882	6.209	1.382	5.083	0.405
4	Broadcast	10.167	0.917	9.792	10.542	0.856	6.073	1.246	5.083	0.313
4	LPG optimal routing	2.000	0.000	2.000	2.000	0.887	6.258	1.431	5.083	0.006
8	Anchor weighted routing	4.875	0.850	4.542	5.230	0.861	6.072	1.204	6.333	0.405
8	Broadcast	12.667	1.090	12.208	13.083	0.855	6.062	1.193	6.333	0.313
8	LPG optimal routing	2.000	0.000	2.000	2.000	0.881	6.214	1.345	6.333	0.006
16	Anchor weighted routing	6.000	0.780	5.667	6.292	0.859	6.063	1.149	7.688	0.405
16	Broadcast	15.375	0.875	15.042	15.708	0.855	6.056	1.143	7.688	0.313
16	LPG optimal routing	2.000	0.000	2.000	2.000	0.877	6.190	1.276	7.688	0.006
32	Anchor weighted routing	5.917	0.654	5.667	6.125	0.859	6.066	1.143	9.521	0.405
32	Broadcast	19.042	1.042	18.625	19.417	0.854	6.050	1.128	9.521	0.313
32	LPG optimal routing	2.000	0.000	2.000	2.000	0.872	6.157	1.234	9.521	0.006

Table S7.2. Scaling exponent summary

protocol	scaling_beta	scaling_beta_ci_low	scaling_beta_ci_high	sublinear
Broadcast	0.313	0.297	0.329	True
Anchor weighted routing	0.405	0.382	0.423	True
LPG optimal routing	0.006	-0.000	0.019	True

S7 interpretation

The GAS curves show that all three protocols can form a shared collective anchor, but they do not do so at the same rate. Broadcast improves monotonically but slowly because each agent receives a dense group summary that must be reconciled against its private anchor. Anchor weighted routing improves faster because it exploits existing overlap, reducing the semantic distance that each update must cover. LPG optimal routing improves fastest because it directly aligns the group to the estimated collective anchor.

The capacity curves track the GAS curves. This is important because it shows that consensus is not only a geometric event in the anchor space. It also increases the effective semantic bandwidth of the group. In practical terms, a network with stronger collective anchor alignment can carry a higher amount of recoverable meaning over the same high SNR physical channel.

The routing gain plot shows that the benefit of semantic routing becomes more visible as N grows. The interpretation is not that broadcast fails, but that broadcast becomes inefficient when the number of simultaneous heterogeneous belief bases grows. LPG optimal routing functions as a semantic scheduling principle: send meaning along routes that reduce anchor distance most efficiently.

S7 implications

Implication for 6G semantic networks. A semantic layer should not treat all nodes as equivalent endpoints. It should estimate anchor compatibility and use that estimate to choose which nodes exchange summaries first. This turns routing from a packet path problem into a meaning alignment problem.

Implication for collaborative AI systems. Multi agent systems can benefit from collective anchor formation before attempting complex joint reasoning. The simulation suggests that pre alignment can raise group capacity and reduce the number of rounds required for shared understanding.

Implication for LPG theory. The results support the claim that the LPG framework extends beyond dyadic transmission. The network does not merely transmit messages; it converges toward a collective semantic state.

Experiment S8: Corrupted anchor robustness

Method

S8 holds the physical channel at 20 dB SNR and degrades the semantic anchor itself. Four components of the sem object were corrupted independently: B, W, Gamma and iota. Corruption intensities were 0, 5, 15, 30, 50 and 80 percent.

Basis corruption replaces concept nodes with out of domain embeddings. Weight corruption perturbs relevance rankings. Gamma corruption perturbs the context action operator. Identity corruption replaces the referential identity tag with a semantically distinct identity mixture. The metrics were C_s, CFI, CAR, SDP, failure thresholds and mixed corruption cascade ratio.

The mixed corruption experiment tested all pairwise component combinations at 15, 30, 50 and 80 percent. Observed semantic loss was compared with the additive expected loss from the corresponding single corruptions. Ratios above 1 indicate superadditive cascade behaviour.

S8 result graphs

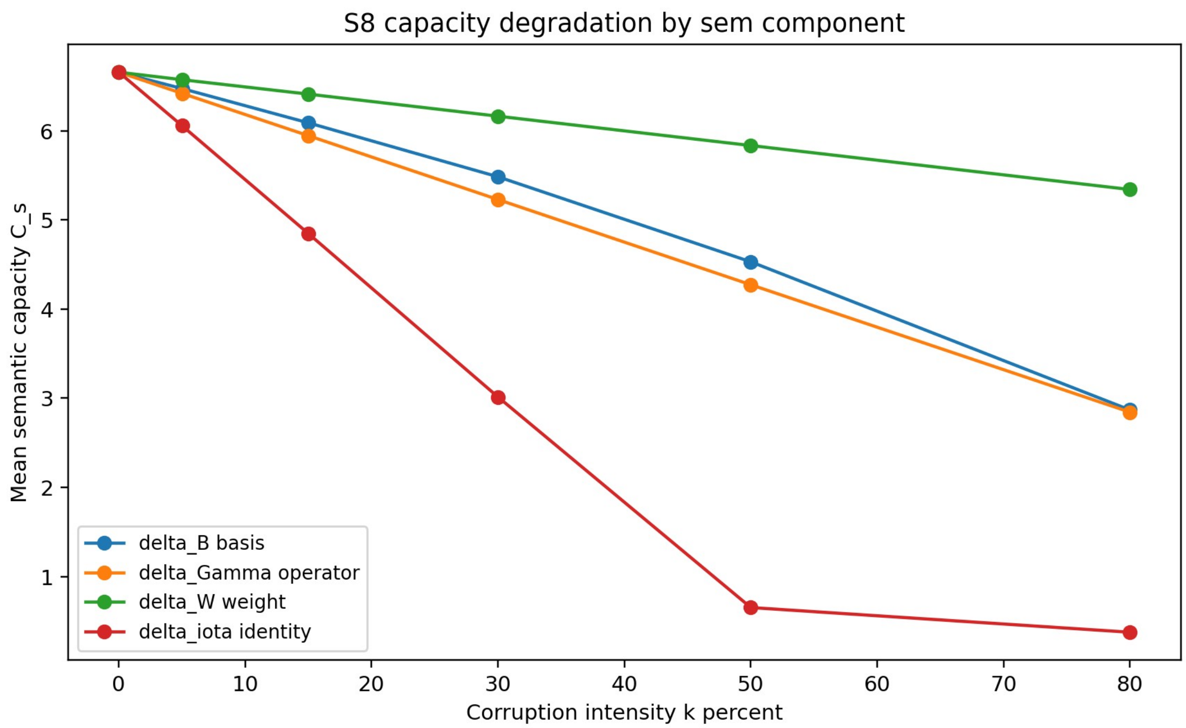

Figure S8.1. Semantic capacity degradation by corrupted sem component.

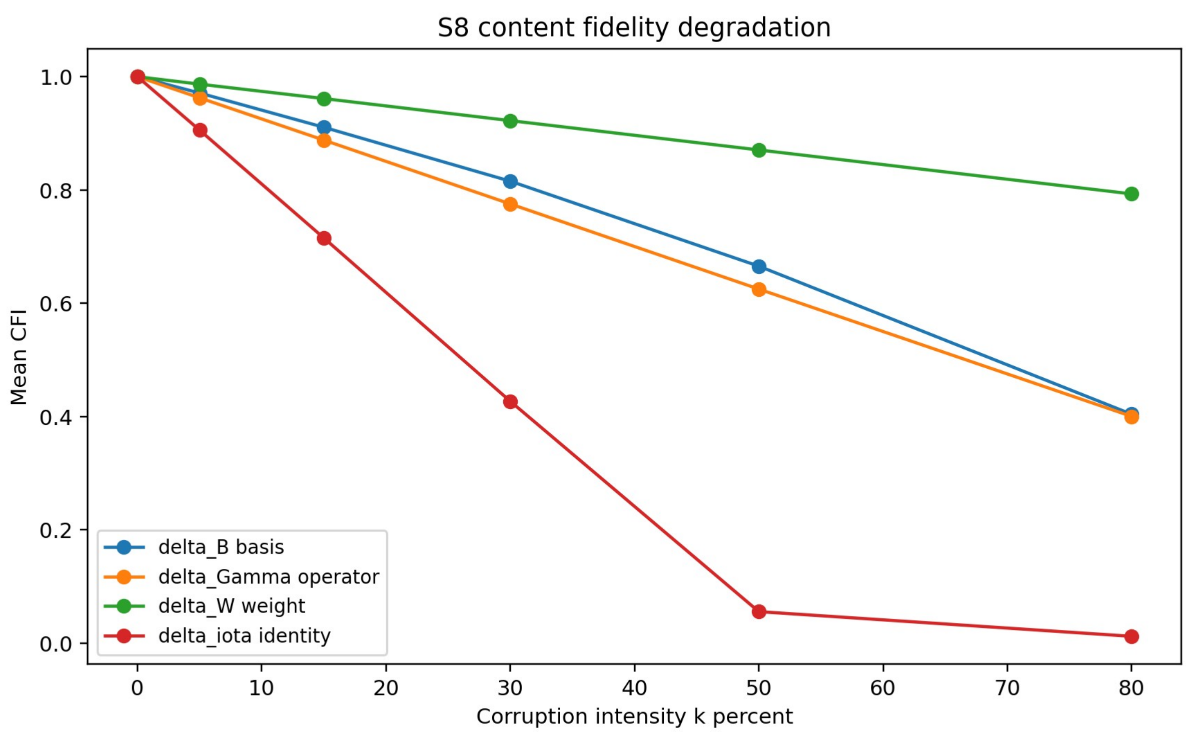

Figure S8.2. Content fidelity degradation by corrupted sem component.

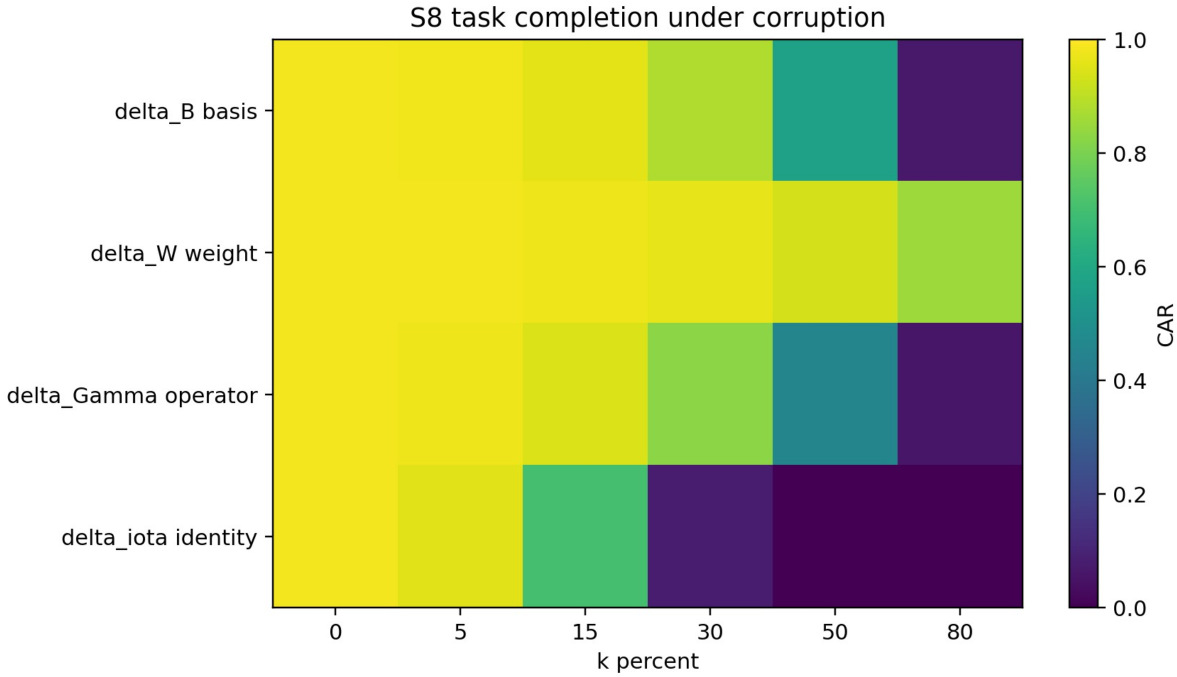

Figure S8.3. Task completion heat map across corruption intensity.

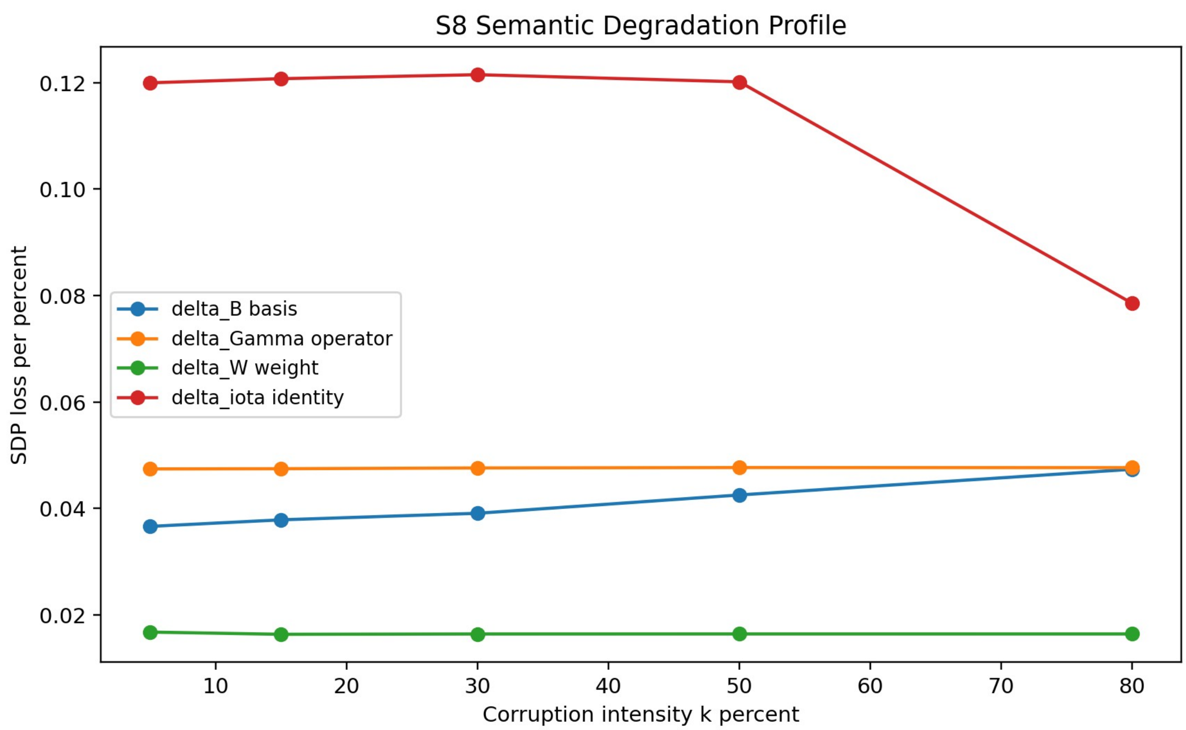

Figure S8.4. Semantic Degradation Profile curves.

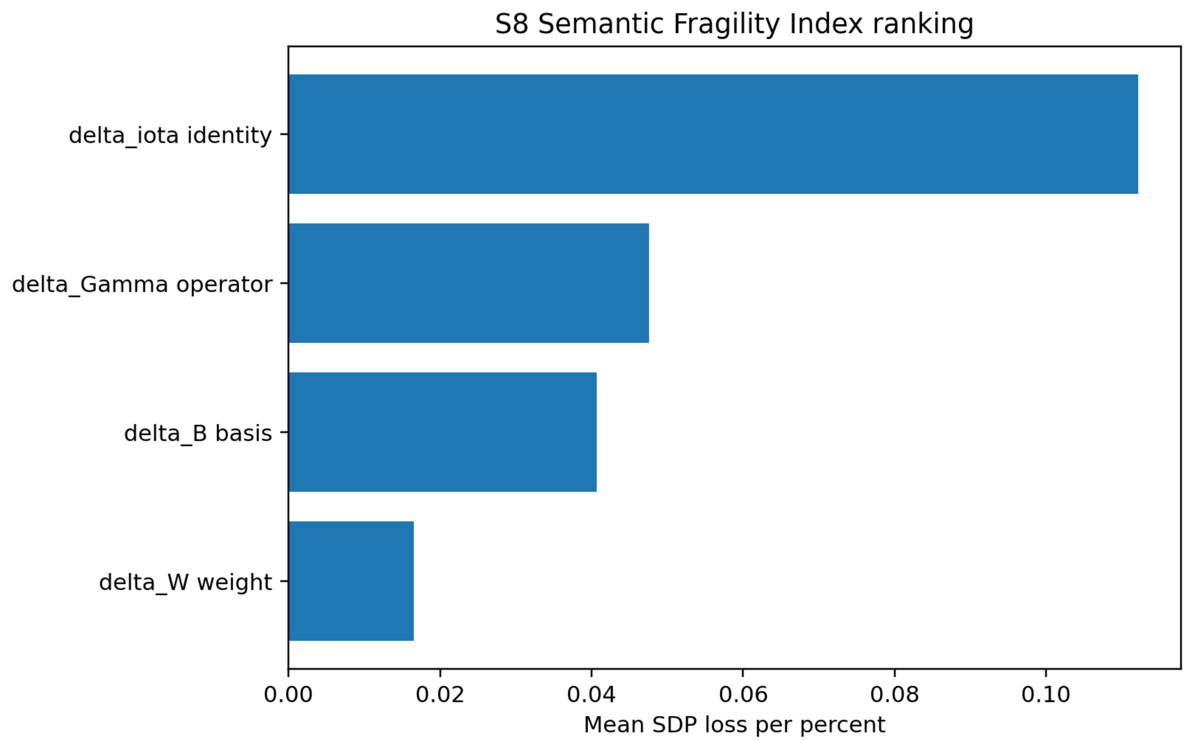

Figure S8.5. Semantic Fragility Index ranking.

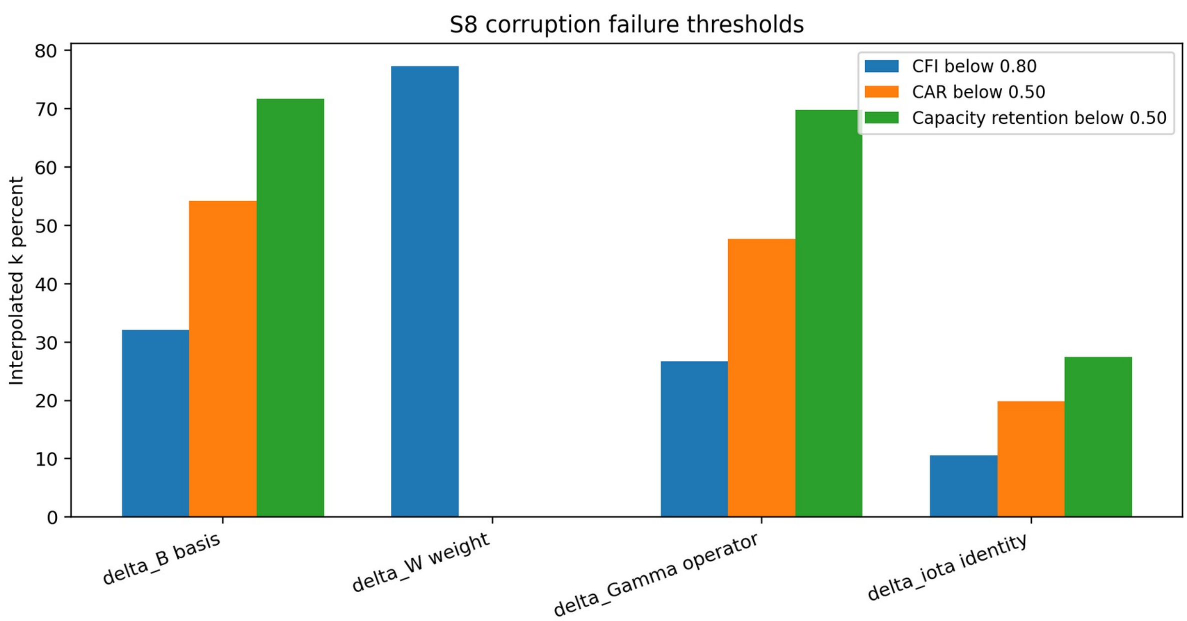

Figure S8.6. Interpolated failure thresholds.

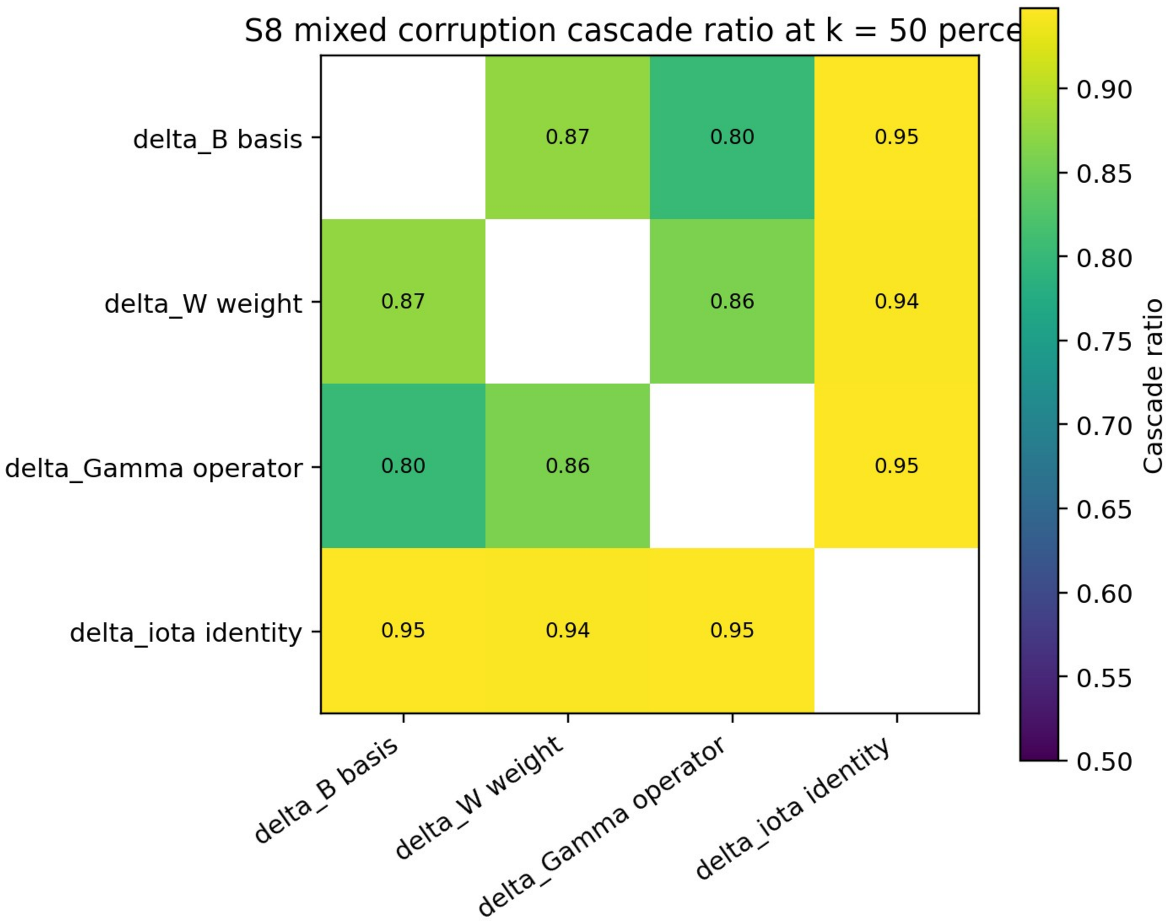

Figure S8.7. Pairwise mixed corruption cascade ratio at k = 50 percent.

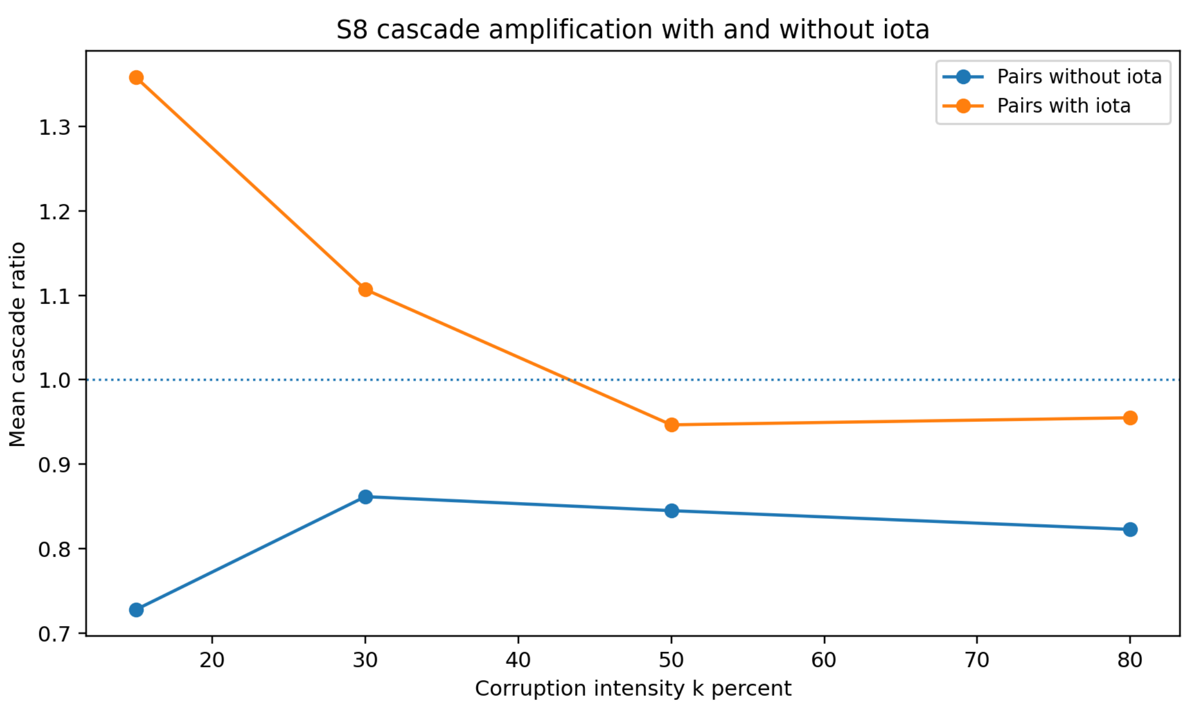

Figure S8.8. Mean cascade ratio for pairs with iota and without iota.

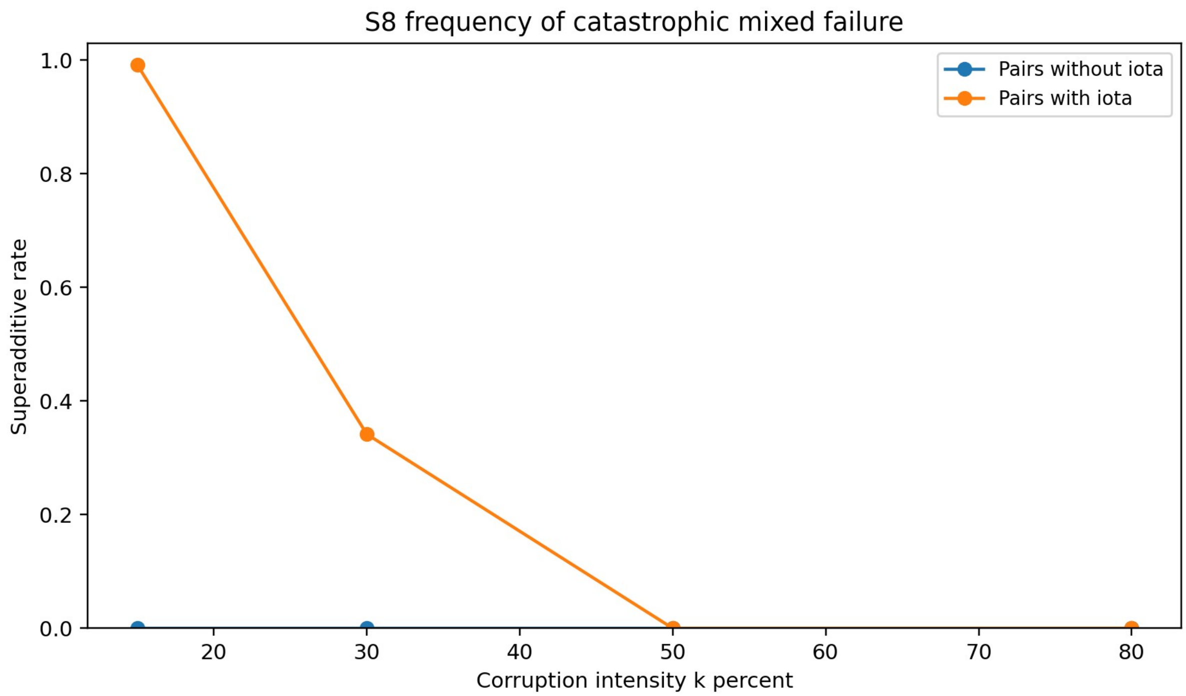

Figure S8.9. Frequency of superadditive mixed corruption failure.

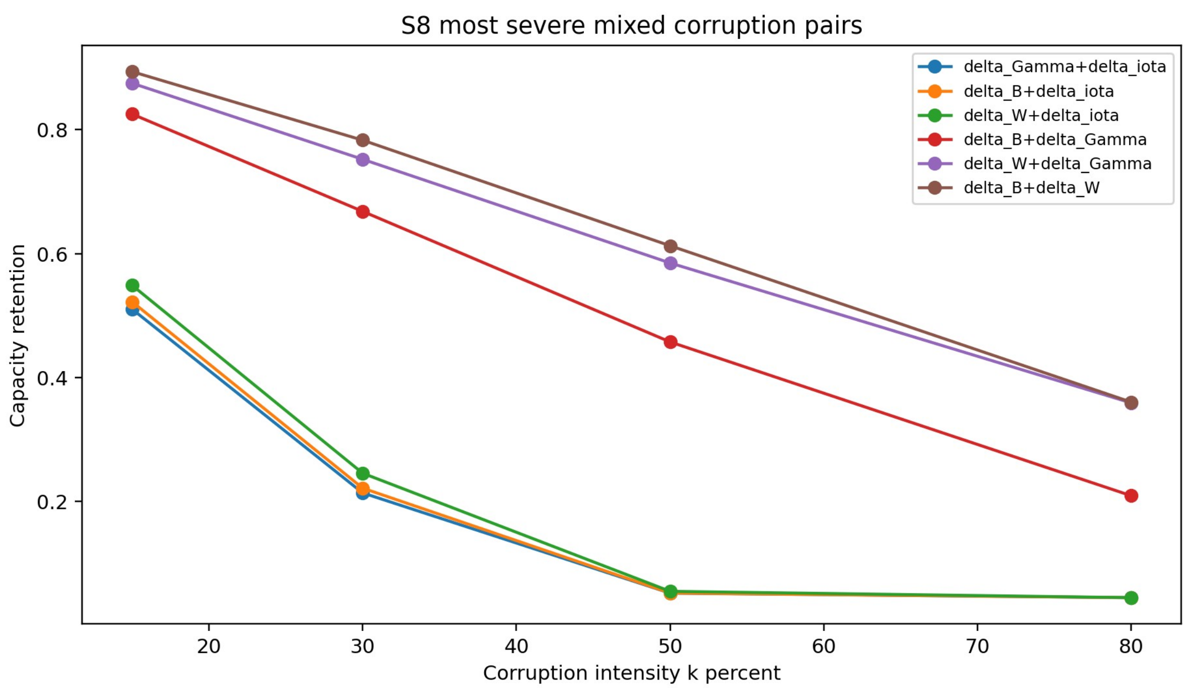

Figure S8.10. Capacity retention for the most severe mixed corruption pairs.

S8 numerical results

Table S8.1. Semantic Fragility Index

rank_fragility	mode	mean_SDP	max_SDP	C_s_at_80	CFI_at_80	CAR_at_80
1	delta_iota identity	0.1122	0.1215	0.3745	0.0118	0.0007
2	delta_Gamma operator	0.0476	0.0477	2.8438	0.4001	0.0596
3	delta_B basis	0.0407	0.0474	2.8680	0.4039	0.0682
4	delta_W weight	0.0165	0.0168	5.3420	0.7930	0.8530

Table S8.2. Failure threshold estimates

mode	k_at_CFI_below_0_80	k_at_CAR_below_0_50	k_at_capacity_retention_below_0_50
delta_B basis	32.061	54.225	71.681
delta_W weight	77.297	nan	nan
delta_Gamma operator	26.713	47.660	69.815
delta_iota identity	10.548	19.894	27.421

Table S8.3. Mixed corruption cascade summary

mode_pair	k_percent	contains_iota	C_s_mean	capacity_retention_mean	CFI_mean	CAR_mean	cascade_ratio_mean	superadditive_rate	cascade_type_dominant
delta_W+delta_iota	15	True	3.655	0.549	0.528	0.215	1.459	1.000	superadditive
delta_B+delta_iota	15	True	3.478	0.522	0.500	0.167	1.325	0.975	superadditive
delta_Gamma+delta_iota	15	True	3.399	0.510	0.487	0.147	1.292	1.000	superadditive
delta_B+delta_Gamma	15	False	5.493	0.825	0.817	0.882	0.894	0.000	additive
delta_W+delta_Gamma	15	False	5.824	0.875	0.869	0.933	0.696	0.000	additive
delta_B+delta_W	15	False	5.948	0.893	0.888	0.945	0.592	0.000	additive
delta_W+delta_iota	30	True	1.637	0.246	0.210	0.007	1.212	1.000	superadditive
delta_B+delta_iota	30	True	1.476	0.222	0.185	0.005	1.077	0.025	additive
delta_Gamma+delta_iota	30	True	1.425	0.214	0.177	0.005	1.031	0.000	additive
delta_B+delta_W	30	False	5.212	0.783	0.773	0.817	0.874	0.000	additive
delta_W+delta_Gamma	30	False	5.007	0.752	0.740	0.760	0.858	0.000	additive
delta_B+delta_Gamma	30	False	4.448	0.668	0.652	0.535	0.852	0.000	additive
delta_Ga	50	True	0.349	0.052	0.008	0.001	0.948	0.000	additive

mma+delta_iota									
delta_B+delta_iota	50	True	0.350	0.053	0.008	0.001	0.947	0.000	additive
delta_W+delta_iota	50	True	0.368	0.055	0.011	0.001	0.945	0.000	additive
delta_B+delta_W	50	False	4.079	0.613	0.594	0.378	0.874	0.000	additive
delta_W+delta_Gamma	50	False	3.894	0.585	0.565	0.297	0.859	0.000	additive
delta_B+delta_Gamma	50	False	3.046	0.457	0.432	0.086	0.801	0.000	subadditive
delta_B+delta_iota	80	True	0.300	0.045	0.000	0.001	0.955	0.000	additive
delta_Gamma+delta_iota	80	True	0.300	0.045	0.000	0.001	0.955	0.000	additive
delta_W+delta_iota	80	True	0.300	0.045	0.000	0.001	0.955	0.000	additive
delta_B+delta_W	80	False	2.399	0.360	0.330	0.030	0.847	0.000	subadditive
delta_W+delta_Gamma	80	False	2.391	0.359	0.329	0.027	0.831	0.000	subadditive
delta_B+delta_Gamma	80	False	1.395	0.209	0.172	0.005	0.791	0.000	subadditive

S8 interpretation

The capacity and CFI curves show a clear hierarchy of fragility. Weight corruption is the most resilient because the weighted concept field can tolerate moderate ranking noise. Basis corruption is more damaging because replacing concept nodes changes the semantic content itself. Gamma corruption is more damaging again because it changes contextual framing. Identity corruption is the most damaging because it changes the referential binding that tells the receiver what the semantic object is about.

The CAR heat map is especially important because it translates capacity loss into task behaviour. In the prototype, identity corruption causes task performance to fall earlier and faster than the other modes. This is the empirical signature expected if iota is not merely metadata but a minimal component of the sem object.

The mixed corruption results show that iota changes the failure regime. Pairs without iota tend to be closer to additive or mildly subadditive behaviour. Pairs with iota show a higher cascade ratio and a higher superadditive rate. This suggests that identity loss removes a stabilising reference frame, making the other corruptions more destructive than they would be separately.

S8 implications

Implication for semantic security. Identity tags require stronger integrity protection than ordinary relevance weights. If an adversary can induce identity collision or identity substitution, the semantic channel can fail even when the physical layer remains clean.

Implication for anchor maintenance. Systems should monitor not only signal SNR but also anchor health. A high physical SNR does not guarantee high semantic capacity if the anchor components are corrupted.

Implication for LPG minimality. The prototype supports the minimality claim for iota. However, because this is synthetic, the claim should be treated as a target for empirical falsification using SNLI and TriviaQA rather than as a final proof.

Combined implications for S7 and S8

S7 and S8 are complementary. S7 asks how meaning becomes shared across a network. S8 asks what happens when the semantic substrate that makes sharing possible is damaged. Together they show that LPG engineering must include both semantic routing and semantic integrity.

A robust LPG based 6G system would therefore need at least three layers: physical reliability, anchor alignment and sem object integrity. Physical reliability keeps symbols intact. Anchor alignment lets agents converge. Sem object integrity prevents the meaning object from losing its basis, weighting, operator or identity.

Recommended next empirical run

For S7, the next run should replace synthetic anchors with domain clustered Wikipedia paragraph embeddings. Each agent should be initialised from a distinct topic cluster, then the same consensus engine should be rerun. The reported output should preserve GAS, T_c, C_s, SRG and scaling beta so the synthetic and empirical results remain comparable.

For S8, the next run should use SNLI for entailment level semantic failure and TriviaQA for identity sensitive factual recall. The same corruption sweep should be performed, but CFI and CAR should be computed from labelled semantic entailment and factual correctness rather than synthetic cosine recovery.

A useful falsification test is to attempt to protect B, W and Gamma while corrupting iota. If the empirical task performance remains stable, the minimality claim for iota must be revised. If task performance collapses, the result strengthens the LPG minimality argument.

Limitations

The values in this report are not final empirical measurements. They are prototype outputs produced by a controlled synthetic generator. They are valid as simulation scaffolding, sensitivity analysis and expected result patterns, but the claims require validation on the external datasets named in the simulation design.

The models also simplify several implementation issues. They do not simulate real transformer summary generation, Neo4j collective anchor graph storage, HuggingFace embedding extraction, Sionna physical layer modelling or Weights and Biases experiment management. Those tools should be added in the full empirical version.

Appendix: output inventory

The deliverable package contains raw CSV files, summary CSV files, graph PNGs, the executable Python script, this DOCX report, a PDF copy and an Excel workbook with result tables. The raw data files allow independent recalculation of all reported metrics.